

Empathy Engine: Using Game Design and Real-time Technology to Cultivate Social Connection

Yuanlinxi Li*
Shenzhen International School of
Design, Harbin Institute of
Technology
China
liyuanlinxi1009@gmail.com

Mengyao Guo*
Shenzhen International School of
Design, Harbin Institute of
Technology
China
University of Macau
China
guomengyao@hit.edu.cn

Ze Gao†
Hong Kong University of Science and
Technology
Hong Kong
zegaoap@hotmail.com

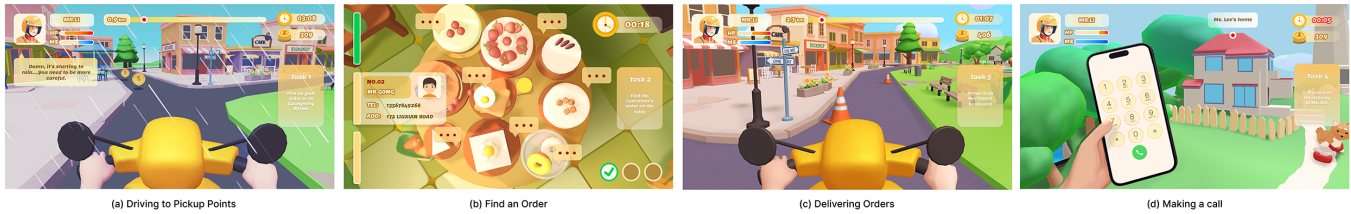


Figure 1: The VR Game of “Takeaway Rider”, Four Main Sections of the Game Process by using Spline Default Models.

Abstract

Our VR game simulates the experiences of takeaway riders, using real-time data to create scenarios that foster empathy between consumers and riders while highlighting the challenges faced by riders. It employs narrative prompts, empathy-based rewards, and assessment scales to measure cognitive and affective empathy improvements. The VR format elicits stronger emotional resonance than 2.5D prototypes, demonstrating the potential of immersive technologies to foster mutual understanding across different social groups.

ACM Reference Format:

Yuanlinxi Li, Mengyao Guo, and Ze Gao. 2024. Empathy Engine: Using Game Design and Real-time Technology to Cultivate Social Connection. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697867>

1 Introduction

In China, takeaway riders are a gig-type employment group on digital platforms. Since the pandemic, their escalating conflicts with consumers, such as “takeaway rider injures customers,” reported by Dahe Daily¹ and Guangming², have gained online prominence.

*Both authors contributed equally to this research (co-first authors).

†Ze Gao is the Corresponding Author.

¹Dahe Daily: <https://baijiahao.baidu.com/s?id=1690733102605832626&wfr=spider&for=pc>

²Guangming.com: <https://m.gmw.cn/baijia/2021-08/29/1302520079.html>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA Posters '24, December 03–06, 2024, Tokyo, Japan

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1138-1/24/12

<https://doi.org/10.1145/3681756.3697867>

From our interviews with consumers ($C = 12$) and takeaway riders ($R = 6$), the negative reviews that takeaway riders trapped in the constrained delivery time are the root cause. Riders must navigate complex and often unpredictable road conditions that can further hinder their ability to deliver on time, piling on the pressure in an already demanding job [Lai 2020]. The evidence gathered from these interviews points to a significant gap in consumer understanding of the realities takeaway riders face.

This game “Takeaway Rider,” demonstrates the potential of interactive, real-time design in engaging and influencing consumer perceptions. Based on the four-stage model of ELT (Concrete Experience, Reflective Observation, Abstract Conceptualization, and Active Experimentation) [Kolb et al. 2014], we built the framework for the game “Takeaway Rider” (see Figure 2). It provides customers (players) with opportunities for authentic experiences, self-reflection, and practical application to enhance their empathic abilities.

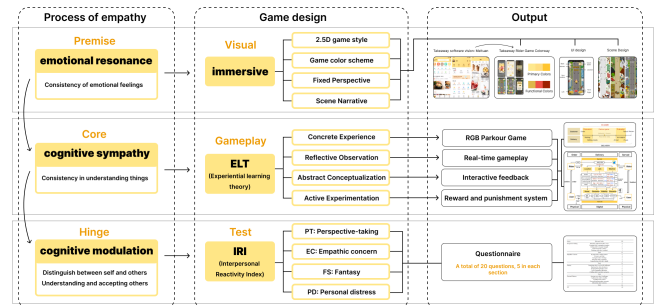


Figure 2: The Theoretical Framework for Empathy in Game Design.

2 Related work

Empathy, the ability to understand others' emotions, is crucial for fostering harmonious relationships and cooperation. However, mobile social networks have reduced the richness of emotional cues typically present in face-to-face interactions. Games offer an alternative by providing immersive role-play experiences that can help develop empathy skills [Belman and Flanagan 2010].

Research on empathy-focused games includes "PeaceMaker," where players act as presidents navigating the Israeli-Palestinian conflict. "Nevermind," developed by Lobel et al., uses biofeedback to adjust difficulty based on heart rate [Lobel et al. 2016]. Similarly, Kors et al. created "A Breathtaking Journey," a mixed reality game simulating refugee experiences [Kors et al. 2016], while Tong et al.'s "AS IF" allows players to experience life with cerebral palsy [Tong et al. 2017].

Based on these innovations, our project addresses the uncertainty about the validity of the findings by providing theoretical models on empathy in games, demonstrating how real-time interactive game design can foster consumer empathy.



Figure 3: The 2.5D Game of "Takeaway Rider" 1.0 by Unity.

3 Approach

"Takeaway Rider" progresses from a fundamental 2.5D prototype to a more engaging version using VR technology. We established the rules by translating real-time weather and traffic data into the game using the AMAP30 SDK development interface (see Figure 3). It integrates AMAP map data, extracting information to determine key game parameters like location, obstacle density, and weather visibility. The system maps real-world data to game metrics, confirming content based on delivery rider order details. It confirms map size for rider delivery time, determining timeslots, environment, and architectural style for a visually consistent representation (see Figure 4). Game difficulty processing logic correlates congestion level with delivery speed impact, influencing obstacle frequency, such as congestion levels (low: 0–30, moderate: 31–60, high: 61–100) define obstacle density (low: 10, moderate: 15, high: 20), while rain or lighter snow adds penalty obstacles like puddles or snowdrifts.

For the second stage, we applied VR for an advanced immersive experience (see Figure 1). We employed the Meta Quest VR headset as the platform for our game with Unity version 2022.3.16f1 and it gave us a framework for testing an immersive environment. When designing the control mechanics, we utilized the XR Interaction Toolkit, a suite of tools available in Unity that simplifies the development of user interactions. With Spline, we designed the game's assets, which included the architecture of the virtual cityscape and the characters that players encounter within the game levels. This enabled us to construct a detailed and visually appealing world that supports the narrative and gameplay.

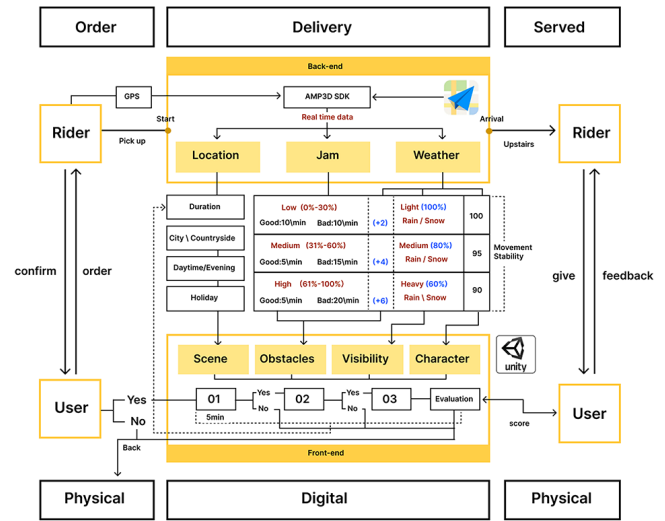


Figure 4: The Game Workflow.

4 Conclusion

This project applied real-time interaction techniques to game design to enhance empathy for delivery riders. It incorporates Experiential Learning Theory (ELT) to provide a structured cognitive and emotional empathy framework. In the future, we will conduct a rigorous user study to design a scale to assess the empathy performance of the VR game model based on the Interpersonal Reactivity Index (IRI) principles [Davis et al. 1996]. Meanwhile, the next step of the study will be to collect real-time scenarios of delivery riders through head-mounted cameras and feed them back in the game scenarios to achieve better empathy.

References

- Jonathan Belman and Mary Flanagan. 2010. Designing games to foster empathy. *International Journal of Cognitive Technology* 15, 1 (2010), 11.
- Mark H Davis et al. 1996. A social psychological approach.
- David A Kolb, Richard E Boyatzis, and Charalampos Mainemelis. 2014. Experiential learning theory: Previous research and new directions. In *Perspectives on thinking, learning, and cognitive styles*. Routledge, 227–247.
- Martijn JL Kors, Gabriele Ferri, Erik D Van Der Spek, Cas Ketel, and Ben AM Schouten. 2016. A breathtaking journey. On the design of an empathy-arousing mixed-reality game. In *Proceedings of the 2016 Annual Symposium on Computer-Human Interaction in Play*. 91–104.
- Youxuan Lai. 2020. Takeaway rider, stuck in the system. <https://mp.weixin.qq.com/s/Mes1RqIdp48CMw4pXTwXw>

Adam Lobel, Marientina Gotsis, Erin Reynolds, Michael Annetta, Rutger CME Engels, and Isabela Granic. 2016. Designing and utilizing biofeedback games for emotion regulation: The case of nevermind. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. 1945–1951.

Xin Tong, Servet Ulas, Weina Jin, Diane Gromala, and Chris Shaw. 2017. The design and evaluation of a body-sensing video game to foster empathy towards chronic pain patients. In *Proceedings of the 11th EAI International Conference on Pervasive Computing Technologies for Healthcare*. 244–250.